

Title

Continuous glucose monitoring systems improve measurement precision and reduce nursing burden in sepsis management: a randomized validation study

Abstract

Background: Accurate real-time glucose monitoring in critical illness enables better insulin titration and avoids hypoglycaemia. We compared a wearable continuous monitoring device against standard fingerstick and laboratory-based glucose measurement in septic patients. **Methods:** Sixty ICU patients with confirmed sepsis were randomized to continuous glucose monitoring (CGM) plus standard care or standard care alone. CGM readings were compared against laboratory values using correlation analysis and Clarke error grid analysis. Protocol adherence, measurement time, and hypoglycaemic episode detection were monitored. **Results:** The CGM system demonstrated excellent agreement with reference methods (correlation coefficient $r=0.986$; $p<0.001$) with mean absolute relative deviation of 11.44%. Clarke error grid analysis showed 91.9% of readings in clinically safe zone A and 8.1% in zone B. The CGM group achieved higher glycaemic protocol adherence (73.91% versus 48.16%; $p<0.001$) and required shorter measurement intervals (41 seconds versus 64 seconds; $p<0.001$). Hypoglycaemic events were detected more frequently with CGM (2.46% versus 0.28%; $p<0.001$). Time within target glucose range was 33.23% in CGM versus 26.26% in standard care. Adverse events were not increased. **Conclusion:** CGM provides accurate real-time glucose monitoring, improves protocol adherence, and enhances early hypoglycaemia recognition in critically ill septic patients.